

🎓 From the Subscriber Mailbag: Cracking the Code on Charm

With Follow up on VANNA



DOC MCGRAW

JAN 10, 2026



14



7



1

Share



🎓 From the Subscriber Mailbag: Cracking the Code on Charm

One of our subscribers dropped a great question in the Feedback Box this morning. He's doing the work—digging into the PDF guide and trying to wrap his head around the nuances of **Charm** (Delta Decay) and 0DTE mechanics.

Doc McGraw - \$SPX Trading / Options Gelt is a reader-supported publication. To receive new posts and support my work, consider becoming a free or paid subscriber.

It's a common sticking point for traders moving from "Technical Analysis" to "Market Structure," so I wanted to share the answer with everyone.

📌 ⌚ Time & Vol Sensitivity (Charm & Vanna)

- **Charm** = delta decay over time
 - Positive → MM buys stock as time passes
- **Vanna** = delta shift from vol moves
 - Implied vol drops → MM buys to rebalance

These effects drive **intraday drift** even without spot moves.

🤔 The Question:

“Doc, I get that Charm (the decay of Delta over time) forces flows late in the day. But why can’t I just sum up the Charm effect at 10:00 AM to see where the market is going? Why does the math only seem to work into the close?”

💡 The Answer:

You can’t just “sum up” Charm early in the day because it gets drowned out. It comes down to a Tug-of-War between two forces: Gamma (Price) vs. Charm (Time).

1. Morning Session: The Rock Concert 🎸

Between 9:30 AM and 1:00 PM, time is moving, but “slowly” relative to expiration.

If SPX moves 5 - 10 points, option Deltas change because of Gamma (sensitivity to price). The change in Delta caused by 1 hour of time passing is tiny compared to a 5-point candle.

- **The Reality:** Gamma dominates. Charm is mathematically there, but it’s like trying to hear a whisper at a rock concert. The price action (Gamma) drowns out the time decay (Charm).

2. Afternoon Session: The Cliff

This is where the 0DTE magic happens (usually 3:00 PM - 4:00 PM).

Charm is the derivative of Theta, and Theta is exponential, not linear. As we approach the bell, the probability of an OTM option going ITM collapses toward zero.

- **The Mechanic:** An option’s Delta isn’t just “shrinking”—it is **crashing** to zero.
- **The Flow:** If a Dealer is Short Calls (Long Delta), and that Delta evaporates simply because the clock ticked, they are suddenly “Over-Long.” They **must** sell futures to flatten the book.

The Bottom Line:

Early in the day, you trade the Levels (Gamma). Late in the day, you trade the Drift (Charm).

*This is the kind of granular, structural edge we build every day in the premium sections. We don't just guess where the line goes; we learn **why** the line moves.*

After I posted this in NOTES, a follow-up question was asked about VANNA (not the game show hostess). So, I wrote up a part 2 and decided to make this Educational Post for the archive.

Here is the follow-up post. It keeps the same “educational drop” format, extending the metaphor to complete the trilogy of forces (Price, Time, Volatility).

Part 2: And Then Vanna Walks In...

Since we're peeling back the curtain on the Greeks today, we have to introduce the third guest at the party.

We established that Gamma is the driver (Price) and Charm is the clock (Time).

So, where does Vanna fit in?

Vanna is the “Mood” (Volatility).

While Gamma watches the chart and Charm watches the clock, **Vanna watches the VIX**. Vanna measures how much an option's Delta changes when Implied Volatility (IV) changes.

Here is how Vanna joins the party and changes the flows:

🔴 The Balloon Analogy

Think of the market's potential range for the day as a balloon.

- **High IV (VIX Spiking):** The balloon inflates. The market thinks price can go anywhere. Far OTM strikes suddenly feel “close.”
- **Low IV (VIX Crushing):** The balloon deflates. The market thinks price is stuck. Even nearby strikes feel “far away.”

1. The “Turbo-Charm” Effect (Vol Crush) 📉

You often see this on a slow, grinding rally or after a news event passes (like 10:00 AM post-data).

- **** The Setup:**** Spot is grinding UP, so IV is melting DOWN.
- **The Mechanic:** As IV drops, the probability of OTM options hitting ITM shrinks rapidly.¹

- **The Result:** Vanna works *with* Charm. It sucks the life out of OTM Deltas faster than time alone would. This is why “Vanna Rallies” feel so frictionless—the board is effectively clearing itself out of resistance as volatility dies.

2. The “Zombie” Effect (Vol Pop) 🧟

This is what happens when we sell off hard or hit an unexpected air pocket.

- **The Setup:** Spot drops, VIX spikes.
- **The Mechanic:** Suddenly, options that Charm had “killed” come back to life. Because IV is expanding, the math says, “*Hey, this 50-point move is actually possible now.*”
- **The Result:** Vanna *fights* Charm. Delta expands on those OTM Puts even though time is running out. This forces Dealers to chase the move lower (selling to hedge the resurrecting deltas), adding fuel to the fire.

The Cheat Sheet:

- **Gamma** moves the market based on **Price**.
- **Charm** moves the market based on **Time**.²
- **Vanna** accelerates or reverses flows based on **Fear**.

QUANTUITION: Analytics combined with Edge from Experience.

Message Doc McGraw

Doc McGraw - \$SPX Trading / Options Gelt is a reader-supported publication. To receive new posts and support my work, consider becoming a free or paid subscriber.



Recommend Doc McGraw - \$SPX Trading / Options Gelt to your readers

Daily SPX options structure and flow analysis. Team coverage of futures, stocks, and breakouts.
QUANTUITION: Math + Experience

Recommend



14 Likes · 1 Restack

[← Previous](#)[Next →](#)

Discussion about this post

[Comments](#) [Restacks](#)

Write a comment...



Juan Busco Jan 10

...

♥ Liked by **Doc McGraw**

excellent article! very informative and helpful!

♥ LIKE (1) 💬 REPLY

📌 SHARE

1 reply by **Doc McGraw**



Teddy Jan 11

...

Amazing scholarly work. What is the best options tool to visualize , by the minute, EOD Charm/ Vanna decay relative to price?

♥ LIKE 💬 REPLY

📌 SHARE

5 more comments...

© 2026 Doc McGraw · [Privacy](#) · [Terms](#) · [Collection notice](#)
[Substack](#) is the home for great culture